

Crisis Connect: A Multimodal Deep Learning Framework for Real-Time Disaster Incident Classification and Emergency Response in Pakistan

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Abstract—Pakistan is among the most disaster-prone countries in the world, experiencing annual floods, earthquakes, fires, and traffic incidents that claim thousands of lives and cause billions of dollars in economic damage. Despite the scale and frequency of these events, no automated AI-powered disaster incident classification system exists that can process real-world citizen reports containing both image and text data. This paper presents Crisis Connect, a multimodal deep learning framework that classifies disaster incidents from citizen-submitted photographs and short text descriptions into four categories: Earthquake, Flood, Fire, and Traffic Incident. The system employs a ResNet-50 convolutional neural network with transfer learning for image classification and a TF-IDF vectorizer combined with a Multi-Layer Perceptron for text classification. A weighted fusion layer combines the outputs of both models at a 70:30 ratio (image to text) to produce a final prediction alongside a severity estimate. Evaluated on a dataset of 2,537 images split 70/15/15 for training, validation, and testing, the CNN achieves 91.03% test accuracy, the NLP component achieves 89.47%, and the fused model achieves 95.07% overall accuracy. The system is deployed as a publicly accessible web application demonstrating real-world applicability. Our results demonstrate that multimodal fusion consistently outperforms single-modality approaches for disaster classification in the Pakistani context.

Index Terms—disaster classification, deep learning, multimodal fusion, ResNet-50, transfer learning, TF-IDF, emergency response, Pakistan, convolutional neural network, natural language processing

I. INTRODUCTION

Pakistan ranks among the top ten countries most vulnerable to climate-related disasters globally [1]. The country experiences devastating floods annually during the monsoon season, with the 2022 floods submerging one-third of the national territory and displacing over 33 million people [2]. Simultaneously, seismic activity along the Eurasian-Indian tectonic plate boundary places northern Pakistan at constant risk of major earthquakes, while urban fires and traffic incidents claim lives daily across metropolitan areas including Lahore, Karachi, and Islamabad.

Despite the severity and regularity of these events, disaster response in Pakistan remains predominantly manual and reactive. Emergency dispatch relies on telephone calls to over-

whelmed operators, incident verification depends on human review of unstructured citizen reports, and resource allocation decisions are made with incomplete situational awareness. The critical window between a disaster occurring and structured emergency response being initiated is often hours rather than minutes.

Recent advances in deep learning have demonstrated strong performance on image-based disaster classification tasks [3], [4]. However, most existing systems suffer from three fundamental limitations in the Pakistani context. First, they rely exclusively on image data, ignoring the rich textual information citizens provide in their reports. Second, they are trained on datasets that do not reflect Pakistani geography, infrastructure, or disaster types. Third, they are not integrated into citizen-facing platforms that can receive reports at scale.

This paper addresses all three limitations. We present Crisis Connect, a multimodal AI framework that processes both image and text inputs from citizen disaster reports, classifies incidents into four actionable categories, and estimates incident severity. Our key contributions are:

- A multimodal fusion architecture combining ResNet-50 image classification with TF-IDF and MLP text classification, achieving 95.07% accuracy on a four-class disaster dataset.
- A curated disaster image dataset of 2,537 images across four categories relevant to Pakistan's emergency landscape.
- A publicly deployed web application demonstrating real-world applicability of the proposed framework.
- Quantitative evidence that weighted multimodal fusion (70% image, 30% text) outperforms single-modality approaches by 0.90 percentage points over the image-only baseline.

The remainder of this paper is organised as follows. Section II reviews related work. Section III describes the dataset, model architecture, and experimental setup. Section IV presents results and analysis. Section V discusses findings and limitations. Section VI concludes the paper.

II. RELATED WORK

A. Image-Based Disaster Classification

Early work on automated disaster classification used traditional computer vision features such as SIFT and HOG descriptors with support vector machine classifiers [7]. The emergence of deep convolutional neural networks dramatically improved classification performance. Alam et al. [3] demonstrated that VGG-16 and ResNet architectures outperformed traditional approaches on disaster image classification tasks. Subsequent work by Nguyen et al. [4] applied deep CNNs to social media images during Hurricane Harvey, achieving over 85% accuracy on binary damage classification.

The AIDER dataset introduced by Kyrkou and Theodoridis [5] provided a standardised benchmark for aerial disaster image classification, containing images of floods, fires, and collapsed buildings. Our work extends this benchmark by adding a traffic incident class and evaluating on ground-level citizen report images rather than aerial imagery.

B. Text-Based Disaster Classification

Social media text has been extensively studied for disaster detection and classification. Imran et al. [8] developed systems for processing crisis-related tweets using machine learning classifiers. TF-IDF vectorization combined with logistic regression and support vector machines has demonstrated competitive performance on short disaster-related text [9]. More recently, transformer-based models such as BERT have been applied to crisis communication classification [10], though their computational requirements limit deployment in resource-constrained environments.

Our work uses TF-IDF with MLP rather than transformer models for three reasons: computational efficiency for deployment on commodity hardware, interpretability of feature weights for disaster-relevant vocabulary, and sufficient performance on short citizen report text (maximum 200 characters).

C. Multimodal Disaster Classification

Multimodal approaches combining image and text have received increasing attention. Chen et al. [11] demonstrated that combining visual and textual features from social media posts improved disaster classification over single-modality baselines. However, existing multimodal systems typically require simultaneous availability of both modalities and are not designed for the citizen reporting context where text descriptions may be brief or absent.

Our weighted fusion approach explicitly handles the relative reliability of each modality through a tunable weight parameter, allowing the system to leverage stronger image evidence while incorporating textual context as a corrective signal.

III. METHODOLOGY

A. Problem Formulation

Given a citizen-submitted disaster report consisting of an image I and a text description T , our goal is to predict a disaster class $\hat{y} \in \mathcal{C}$ and a severity level $s \in \{\text{Low, Medium, High}\}$, where $\mathcal{C} = \{\text{Earthquake, Flood, Fire, Traffic Incident}\}$.

The fused prediction is defined as:

$$\hat{y} = \arg \max_c [\alpha \cdot P_{\text{CNN}}(c|I) + (1 - \alpha) \cdot P_{\text{MLP}}(c|T)] \quad (1)$$

where $\alpha = 0.70$ is the image weight, P_{CNN} is the probability vector from the CNN, and P_{MLP} is the probability vector from the MLP classifier.

B. Dataset

1) *Image Dataset*: We constructed a dataset of 2,537 images from two publicly available sources:

- **AIDER Dataset [5]**: Aerial image dataset for emergency response containing images of floods, fires, and collapsed buildings.
- **Kaggle Disaster Recognition Dataset**: Approximately 7,000 images across flood, earthquake, wildfire, and hurricane categories.

Images were mapped to our four target classes as follows: flood images from both sources form the Flood class; earthquake damage images form the Earthquake class; fire and wildfire images form the Fire class; road accident and vehicle collision images form the Traffic Incident class.

All images were resized to 224×224 pixels and normalised using ImageNet mean $\mu = [0.485, 0.456, 0.406]$ and standard deviation $\sigma = [0.229, 0.224, 0.225]$ for compatibility with pretrained ResNet-50 weights.

2) *Data Preprocessing*: Preprocessing involved three steps:

- 1) **Corruption removal**: PIL image verification was applied to remove unreadable files.
- 2) **Deduplication**: Perceptual hashing with a similarity threshold of 5 was used to remove near-duplicate images.
- 3) **Class balancing**: Classes were capped at equal sample counts to prevent training bias.

The final dataset was split into training (70%), validation (15%), and test (15%) sets using stratified sampling. Table I shows the resulting distribution.

TABLE I
DATASET DISTRIBUTION ACROSS CLASSES AND SPLITS

Class	Train	Val	Test	Total
Earthquake	397	104	122	623
Flood	482	135	130	747
Fire	438	124	120	682
Traffic Incident	339	72	74	485
Total	1656	435	446	2537

3) *Text Dataset*: A text dataset of 124 samples was constructed: 100 synthetic English descriptions (25 per class) and 24 real Urdu/code-switched audio recordings transcribed using OpenAI Whisper and translated to English using Google Translate, simulating citizen disaster reports with a maximum length of 200 characters. Descriptions were written to reflect realistic reporting language used in Pakistani emergency contexts. Text preprocessing included lowercasing, punctuation removal, and stopword filtering using NLTK, preserving disaster-critical terms such as *not*, *near*, *above*, and *below*.

C. CNN Image Classifier

We employed ResNet-50 [6] pretrained on ImageNet as the image classification backbone. Transfer learning was applied by:

- 1) Freezing all convolutional layers to preserve learned visual features.
- 2) Replacing the final fully connected layer (2048×1000) with a custom head: Dropout($p = 0.4$) followed by Linear(2048×4).

Only the custom head (approximately 8,196 parameters) was trained, compared to the 23 million total parameters in the network. This approach leverages ImageNet visual representations while adapting the classification boundary to disaster-specific categories.

Training configuration:

- Optimiser: Adam with learning rate $\eta = 0.001$
- Learning rate scheduler: StepLR with step size 3 and $\gamma = 0.5$
- Early stopping: patience of 4 epochs on validation loss
- Maximum epochs: 15
- Batch size: 32

Data augmentation applied during training:

- Random resized crop to 224×224
- Random horizontal flip
- Random rotation (± 15)
- Colour jitter (brightness, contrast, saturation)

D. NLP Text Classifier

Text descriptions are preprocessed and vectorised using TF-IDF with unigrams and bigrams, retaining the top 3,000 features by term frequency. A Multi-Layer Perceptron classifier with two hidden layers (512 and 256 neurons, ReLU activation) is trained on the resulting feature vectors.

MLP configuration:

- Architecture: $3000 \rightarrow 512 \rightarrow 256 \rightarrow 4$
- Activation: ReLU
- Regularisation: L2 penalty ($\alpha = 0.001$)
- Optimiser: Adam
- Early stopping: patience 10 on validation score

E. Fusion Layer and Severity Estimation

The fusion layer implements Equation 1 with $\alpha = 0.70$. This weight was selected based on the significantly higher accuracy of the CNN component relative to the MLP component, reflecting the greater informativeness of visual evidence for disaster type identification.

Severity estimation maps the fused confidence score and predicted class to a three-level scale:

$$s = \begin{cases} \text{High} & \text{if } \hat{p} \geq 0.75 \text{ and } \hat{y} \in \{\text{Earthquake, Flood}\} \\ \text{Medium} & \text{if } \hat{p} \geq 0.75 \text{ or } \hat{p} \geq 0.50 \\ \text{Low} & \text{otherwise} \end{cases} \quad (2)$$

where $\hat{p} = \max_c P_{\text{fused}}(c)$ is the maximum fused probability.

F. System Architecture

The end-to-end system consists of:

- **Frontend:** Streamlit web application accepting image upload and text description input, displaying prediction results with class probabilities and severity level.
- **Backend:** FastAPI REST API serving the fusion model at a `/predict` endpoint.
- **Deployment:** Streamlit Cloud with model weights hosted on Google Drive and downloaded at runtime via `gdown`.

IV. RESULTS

A. CNN Classification Results

The ResNet-50 model trained for 15 epochs achieved 91.03% test accuracy. Table II presents the per-class precision, recall, and F1 score.

TABLE II
CNN IMAGE CLASSIFIER — PER-CLASS PERFORMANCE

Class	Precision	Recall	F1	Support
Earthquake	0.893	0.885	0.889	122
Flood	0.982	0.908	0.944	130
Fire	0.946	0.939	0.942	120
Traffic Incident	0.788	0.905	0.843	74
Weighted avg	0.914	0.910	0.911	446

B. NLP Classification Results

The TF-IDF + MLP classifier achieved 89.47% test accuracy on the synthetic text dataset. Performance on the small test set (15 samples) reflects the limited size of the text training corpus.

TABLE III
NLP TEXT CLASSIFIER — PER-CLASS PERFORMANCE

Class	Precision	Recall	F1	Support
Earthquake	0.800	0.800	0.800	5
Flood	0.833	1.000	0.909	5
Fire	1.000	0.800	0.889	5
Traffic Incident	1.000	1.000	1.000	4
Weighted avg	0.904	0.895	0.894	19

C. Fusion Results

The weighted fusion model achieved 95.07% test accuracy on the full 446-sample test set. Table IV presents per-class results.

D. Comparison of Modalities

Table V compares the three approaches.

The fusion model improves over the CNN-only baseline by 0.90 percentage points and over the NLP-only baseline by 16.14 percentage points. The improvement is most pronounced for the Traffic Incident class, where text context provides the strongest corrective signal for visually ambiguous scenes.

TABLE IV
FUSION MODEL — PER-CLASS PERFORMANCE ON TEST SET

Class	Precision	Recall	F1	Support
Earthquake	0.934	0.934	0.934	122
Flood	1.000	0.958	0.979	130
Fire	0.954	0.958	0.956	120
Traffic Incident	0.850	0.919	0.883	74
Weighted avg	0.944	0.942	0.942	446

TABLE V
ACCURACY COMPARISON ACROSS MODALITIES

Model	Test Accuracy
CNN only (ResNet-50)	91.03%
NLP only (TF-IDF + MLP)	89.47%
Fused (70% CNN + 30% NLP)	95.07%

V. DISCUSSION

A. Why Fusion Improves Performance

The improvement from fusion over CNN-alone is consistent with the intuition that text descriptions provide contextual information not always visible in images. A photograph of a flooded road can be ambiguous — it may resemble a waterlogged street after heavy rain. A citizen’s text description stating “water level rising fast near the bridge” disambiguates the scene. The 30% text weight is sufficient to shift borderline CNN predictions toward the correct class without overwhelming visual evidence.

B. Traffic Incident Classification

Traffic Incident consistently shows the lowest precision (0.850) across all three models. This is expected: traffic accident scenes frequently overlap visually with other disaster types. An overturned vehicle near a building may be misclassified as Building Collapse; smoke from a vehicle fire may be misclassified as a structural fire. Future work should augment the Traffic Incident training set with more diverse examples.

C. Text Model Limitations

The 89.47% NLP accuracy reflects the limited size of the text training corpus (100 synthetic descriptions). Real citizen reports in Pakistani contexts would also frequently contain Urdu text, code-switching between Urdu and English, and regional dialect vocabulary. The current system processes only English text. Future versions should incorporate multilingual support using Urdu-capable models or translation preprocessing via Whisper speech recognition and deep-translator.

D. Deployment Considerations

The current deployment downloads the 90MB ResNet-50 model file from Google Drive at startup, introducing a cold-start latency of approximately 30 seconds. Production deployment should cache the model in persistent storage. The FastAPI backend is designed to be deployed independently for higher-throughput environments.

E. Limitations

- The text training corpus is synthetic and small (100 samples). Real-world performance on citizen-authored reports may differ.
- Severity estimation is rule-based and does not account for compound factors such as population density or infrastructure damage extent.
- The system processes static images only. Real-time video classification is out of scope for this version.
- Urdu, Pashto, and Sindhi language inputs are not supported in the current text pipeline.

VI. CONCLUSION

This paper presented Crisis Connect, a multimodal deep learning framework for disaster incident classification in Pakistan. We demonstrated that combining ResNet-50 image classification with TF-IDF and MLP text classification through weighted fusion achieves 95.07% accuracy on a four-class disaster dataset, outperforming single-modality image classification by 0.90 percentage points. The system is deployed as a publicly accessible web application at <https://crisis-connect-vjmeaydb54qdg7y3m3xfes.streamlit.app/>.

The practical contribution of this work extends beyond classification accuracy. Crisis Connect demonstrates that multimodal citizen report processing is both technically feasible and deployable on commodity infrastructure. The framework is designed to be extensible: future work will incorporate Urdu language support via Whisper and deep-translator, expand the training dataset with real citizen reports, and integrate the classification pipeline with the Crisis Connect mobile app backend for automated incident tagging and responder dispatch.

The source code, trained models, and dataset documentation are publicly available at <https://github.com/crisis-connect>.

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